

Supplementary Sensitivity Analyses for Positional-Parameter Robustness

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1. Seed alignment and Welch sensitivity

The ViT-B/16 and ViT-S/16 cohorts use the same six seed labels, but the architectures consume different amounts of the global random stream during model initialization. The training DataLoader does not use a separately locked generator, so equal labels do not establish identical initial weights or data order. Main-text intervals are seed-aligned difference intervals retained for continuity with the within-architecture analyses. Table S1 gives unpaired Welch intervals as a sensitivity analysis.

Table S1: Seed-aligned and unpaired Welch 95% intervals for ViT-S minus ViT-B nAUC contrasts. Both interval constructions give the same interpretive conclusion for every family and metric.

PE family	Metric	Mean difference	Seed-aligned 95% CI	Welch 95% CI
Learned	Adversarial nAUC	-0.218610	[-0.292924, -0.144295]	[-0.293806, -0.143413]
Sinusoidal	Adversarial nAUC	-0.159924	[-0.184240, -0.135608]	[-0.184361, -0.135487]
RoPE	Adversarial nAUC	-0.093204	[-0.160354, -0.026054]	[-0.151529, -0.034878]
Learned	Noise nAUC	-0.000894	[-0.003112, 0.001324]	[-0.002435, 0.000646]
Sinusoidal	Noise nAUC	0.000729	[-0.001757, 0.003216]	[-0.001043, 0.002501]
RoPE	Noise nAUC	0.000132	[-0.000479, 0.000743]	[-0.000417, 0.000681]

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2. Ordinal robustness of the architecture-by-PE interaction

Let Δ_f denote ViT-S minus ViT-B adversarial nAUC for family f . More negative values indicate a larger loss under the smaller backbone. Five of the six seeds have the order Learned < Sinusoidal < RoPE; seed 42 is the sole exception. Table S2 lists the six seed-level orderings explicitly.

Table S2: Per-seed ordering of adversarial architecture effects.

Seed	Learned Δ	Sinusoidal Δ	RoPE Δ	Ascending order
42	-0.121191	-0.159980	-0.153736	Sinusoidal < RoPE < Learned
123	-0.188993	-0.131088	-0.065368	Learned < Sinusoidal < RoPE
456	-0.200771	-0.197074	-0.157989	Learned < Sinusoidal < RoPE
789	-0.205311	-0.174361	0.014230	Learned < Sinusoidal < RoPE
1011	-0.326862	-0.149978	-0.087133	Learned < Sinusoidal < RoPE
1213	-0.268529	-0.147064	-0.109227	Learned < Sinusoidal < RoPE

The rank sums are 8, 11, and 17 for Learned, Sinusoidal, and RoPE, respectively, reproducing the exact Friedman result $Q = 7.0$, $p = 0.0289352$. Table S3 reports the corresponding pairwise difference-in-differences contrasts.

Table S3: Pairwise differences in architecture effects. Same-sign counts refer to the dominant sign across the six seeds.

Difference-in-differences contrast	95% CI	Same-sign seeds
Learned – RoPE	$[-0.235330, -0.015482]$	5/6 negative
Sinusoidal – RoPE	$[-0.133297, -0.000143]$	6/6 negative
Learned – Sinusoidal	$[-0.141594, 0.024222]$	5/6 negative

3. Architecture-dependent seed dispersion

Table S4 reports the seed-level dispersion of the AMP-matched architecture effects.

Table S4: Seed dispersion of adversarial nAUC. The leave-one-out range is the minimum and maximum ViT-S/ViT-B SD ratio after omitting each seed in turn.

PE family	ViT-B range	ViT-S range	ViT-B SD	ViT-S SD	SD ratio	LOO SD-ratio range
Learned	0.007196	0.209900	0.002436	0.071663	29.414	[19.930, 47.272]
RoPE	0.035298	0.149260	0.011945	0.055722	4.665	[3.919, 5.754]
Sinusoidal	0.051913	0.059656	0.018077	0.019832	1.097	[0.614, 1.485]

The dispersion ordering, Learned $\gg$ RoPE $>$ Sinusoidal, differs from the mean architecture-effect ordering, Learned $>$ Sinusoidal $>$ RoPE. This is reported descriptively; no mechanism or universal variance-scaling law is inferred.

4. Noise sign balance and normalized values above one

Table S5 gives the seed-level sign balance for the noise architecture contrasts.

Table S5: Seed signs for ViT-S minus ViT-B nAUC contrasts.

PE family	Attack $\Delta < 0$	Attack $\Delta > 0$	Noise $\Delta < 0$	Noise $\Delta > 0$
Learned	6	0	3	3
Sinusoidal	6	0	2	4
RoPE	5	1	3	3

Noise contrasts have no common sign tendency, unlike the adversarial contrasts. ViT-S Sinusoidal seed 123 has noise nAUC 1.0009374. Normalized accuracy is defined relative to clean accuracy on the same finite split and is not clipped at one; a perturbed model can therefore marginally outperform its clean reference, and its integrated normalized curve can slightly exceed one.

5. Sensitivity to the adversarial lower envelope

The primary analysis retains the highest-loss restart at each native budget and then applies a cumulative lower envelope to adversarial normalized accuracy after ordering points by achieved ρ_{rel} . As a deterministic sensitivity analysis, we keep the same selected restart at every budget but connect the raw measured normalized accuracies directly, using the same exact clean

anchor, achieved-displacement ordering, piecewise-linear interpolation, and no clipping above one.

Table S6: Deterministic sensitivity to omitting the adversarial cumulative lower envelope. Values are family-mean noise-minus-adversarial nAUC gaps across six seeds. “Raw” connects the same loss-selected native-budget points without enforcing monotonicity; Δ is raw minus envelope. Every raw seed-level gap is positive at both endpoints.

Dataset	PE family	$\rho_{\max} = 0.09$			$\rho_{\max} = 0.095$			Raw gap > 0
		Envelope gap	Raw gap	Δ	Envelope gap	Raw gap	Δ	
ImageNet-100	Learned	0.004501	0.004466	-0.000036	0.004802	0.004768	-0.000034	6/6
	Sinusoidal	0.137345	0.137345	+0.000000	0.166796	0.166794	-0.000001	6/6
	RoPE	0.047449	0.047397	-0.000053	0.056335	0.056285	-0.000050	6/6
	ALiBi	0.126587	0.126587	+0.000000	0.140466	0.140466	+0.000000	6/6
CIFAR-100	Learned	0.015369	0.014591	-0.000778	0.018347	0.017030	-0.001317	6/6
	Sinusoidal	0.040536	0.040536	+0.000000	0.044829	0.044829	+0.000000	6/6
	RoPE	0.008677	0.008540	-0.000138	0.010009	0.009879	-0.000130	6/6
	ALiBi	0.020276	0.020276	+0.000000	0.022697	0.022697	+0.000000	6/6

At both integration endpoints, all 48 raw noise-minus-adversarial seed-level gaps remain positive and every within-seed family rank configuration is preserved. Consequently, the exact Friedman results are unchanged: $Q = 16.4$, exact $p = 8.03755 \times 10^{-6}$ for ImageNet-100 and $Q = 15.0$, exact $p = 9.46924 \times 10^{-5}$ for CIFAR-100. The largest change at the primary endpoint is CIFAR-100 Learned PE, whose family-mean gap changes from 0.015369 with the envelope to 0.014591 without it. At $\rho_{\max} = 0.095$, the largest change is again CIFAR-100 Learned PE, from 0.018347 to 0.017030. Thus, the positive-gap and between-family heterogeneity conclusions do not depend on the lower-envelope operation. Because the no-envelope analysis connects the same selected points without a cumulative minimum, it also removes the family-dependent opportunity for envelope selection that can arise when curves contain unequal numbers of measured adversarial budgets.

6. Sensitivity to the post-lock task-loss budget

The canonical ViT-S/16 Sinusoidal task-loss curves were defined on the prespecified native-budget grid ending at 0.010. A historical targeted coverage extension subsequently attempted to reach $\rho_{\text{rel}} = 0.09$ for seeds 123, 456, and 1213 using additional native budgets through 0.020; the run ledger records all three as target-not-reached and the registered combined maxima equal their canonical maxima to ledger precision. Original row-level coverage-extension JSONs are not redistributed. The later all-restart audit is a separate public

row-level audit that used the same checkpoints, split, float32 tensors/model state without autocast or gradient scaling, task-loss objective, and 200×5 schedule, and additionally evaluated native budget 0.020 for all six seeds. Because that point was collected after the canonical grid was locked, it is not inserted into the primary estimand. The analogous ViT-B/16 budget-0.015 audit points reach $\rho_{\text{rel}} \in [0.096597, 0.106636]$, above both the locked ViT-B/16 endpoint and the 0.095 sensitivity endpoint, so they cannot alter the reported ViT-B/16 nAUC values.

As a transparent post-lock sensitivity, we add the loss-selected 0.020 point to each of the six ViT-S/16 Sinusoidal curves and retain the same cross-architecture integration endpoint $\rho_{\text{rel}} = 0.0792384$.

Table S7: Post-lock sensitivity to adding the loss-selected ViT-S/16 Sinusoidal task-loss point at native budget 0.020. The primary column uses the prespecified canonical grid ending at 0.010; the sensitivity column adds the later all-restart audit point while keeping the locked cross-architecture endpoint $\rho_{\text{rel}} = 0.0792384$.

Seed	Audit ρ_{rel}	Audit normalized accuracy	Primary adversarial nAUC	With 0.020 point	Difference
42	0.079453	0.021556	0.796951	0.790104	-0.006847
123	0.071233	0.026967	0.779643	0.732882	-0.046760
456	0.068338	0.018950	0.725753	0.687994	-0.037759
789	0.066990	0.025229	0.812803	0.702306	-0.110496
1011	0.076631	0.020154	0.766697	0.752726	-0.013971
1213	0.067808	0.026297	0.716350	0.664651	-0.051699
Across-seed mean $\pm$ sample SD			0.766366 ± 0.038521	0.721777 ± 0.045899	-0.044589

The added point lowers adversarial nAUC in every seed. The across-seed mean changes from 0.766366 ± 0.038521 to 0.721777 ± 0.045899 , a difference of -0.044589 . This increases the noise-minus-adversarial separation and leaves the adversarial family ordering unchanged. The sensitivity therefore strengthens the central decoupling result, while the prelocked-grid values remain the primary reported estimates.